PART A

Experiment 8

A.1 : Aim

To study and implement automated software testing using **Postman for REST API testing** and **Selenium WebDriver for web application testing** (to understand how automation improves the efficiency, accuracy, and repeatability of software testing.).

A.2: Prerequisites

Students should have:

- Basic knowledge of **Software Testing and SDLC**.
- Basic understanding of **manual testing concepts**, test cases, test scenarios, and test execution.
- Basic knowledge of **HTTP/HTTPS and REST APIs**.
- Basic knowledge of **HTML, web pages, and web browsers**.
- Basic programming knowledge in any one language, preferably **Java/Python**. ● Familiarity with using a computer, browser, and basic command-line operations.

A.3: Lab Outcomes

After successful completion of this laboratory, students will be able to:

1. **Understand** the concepts and importance of automated software testing.
2. **Create and execute API test cases** using Postman for GET, POST, PUT, and DELETE requests.
3. **Validate API responses** using status codes, response data, headers, and assertions. 4. **Develop automated test scripts** for web applications using Selenium WebDriver. 5. **Identify and interact with web elements** such as buttons, text boxes, links, dropdowns, and checkboxes using Selenium.
6. **Execute automated test cases** and analyze the test results.
7. **Compare manual and automated testing** in terms of effort, accuracy, repeatability, and execution time.

A.4 :Theory:

Automated Software Testing

Automated software testing is a software testing technique in which testing tools and scripts are used to execute test cases, compare actual results with expected results, and identify errors in a software application. It reduces manual effort and improves the speed, accuracy, and repeatability of testing. Automated testing is widely used in software development and DevOps to maintain software quality.

Postman for REST API Testing

Postman is a tool used to create, send, and test HTTP requests to REST APIs. It allows testers to inspect response data, status codes, headers, and execution time. Postman supports automated test scripts using JavaScript to verify whether an API response meets the expected conditions.

The main HTTP methods used in API testing are:

- GET: Retrieves data from a server.
- POST: Sends data to create a new resource.
- PUT: Updates or replaces an existing resource.
- DELETE: Requests deletion of a resource.

Postman provides assertions to validate API responses. For example, a tester can check whether the response status code is `200 OK` or whether the returned JSON data contains the expected values. The test results help identify whether the API is working correctly.

Selenium WebDriver for Web Application Testing

Selenium WebDriver is an open-source tool used to automate web browsers for testing web applications. It supports programming languages such as Python, Java, and C#. It allows testers to interact with web elements and verify the behaviour of a website.

Selenium can perform operations such as:

- Opening a website in a browser.
- Entering text into text boxes.
- Clicking buttons and links.
- Selecting dropdown options and checkboxes.
- Verifying page content and confirmation messages.

Selenium WebDriver executes test scripts in a browser and helps identify functional errors. It can also be integrated with testing frameworks to execute multiple test cases automatically.

Advantages of Automated Testing

- Saves time: Repeated test cases can be executed quickly.
- Improves accuracy: Reduces errors caused by repetitive manual testing. ● Ensures repeatability: The same test cases can be executed multiple times. ● Improves test coverage: A large number of test cases can be tested efficiently.
- Supports continuous testing: Automated tests can be integrated into CI/CD pipelines.

- Reduces long-term effort: Frequently repeated tests require less manual work.

Manual Testing vs Automated Testing

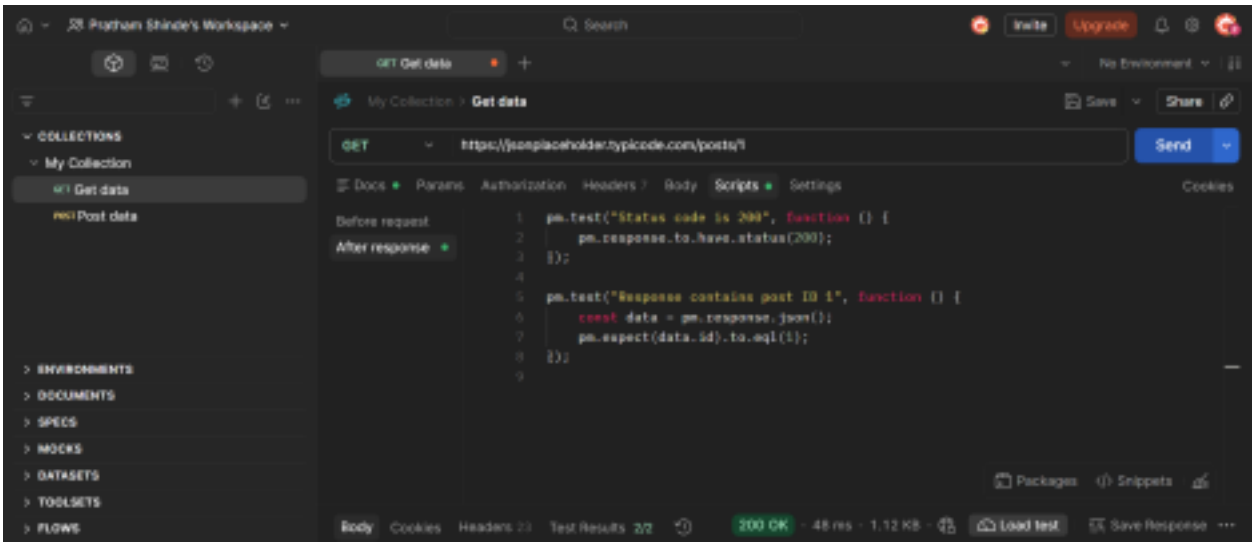
Manual Testing	Automated Testing
Tests are performed manually by a tester.	Tests are executed using tools and scripts.
Repeated testing takes more time.	Repeated testing is faster.
Human errors may occur during repetitive tasks.	Scripted checks provide consistent results.
Suitable for exploratory and usability testing.	Suitable for repetitive and regression testing.

Postman and Selenium serve different purposes. Postman is primarily used for REST API testing, whereas Selenium WebDriver is used for browser-based web application testing. Using both tools helps verify the functionality of APIs and web applications more efficiently.

PART B
(PART B: TO BE COMPLETED BY STUDENTS)

Roll No. A27	Name: Prajwal Ashok Zolage
Class: BE–AI&DS	Batch: A1
Date of Experiment: 06/10/2026	Date of Submission: 06/10/2026

B.1 Code Screenshot:
Postman GET test script under Scripts → After response.



Selenium Python code.

Welcome | A27 exp8.py

A27 exp8.py > ...

```python
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

driver = webdriver.Chrome()

try:
    driver.get(
        "https://www.selenium.dev/selenium/web/web-form.html"
    )

    wait = WebDriverWait(driver, 10)

    text_box = wait.until(
        EC.visibility_of_element_located((By.NAME, "my-text"))
    )
    text_box.send_keys("Automated Testing")

    driver.find_element(By.TAG_NAME, "button").click()

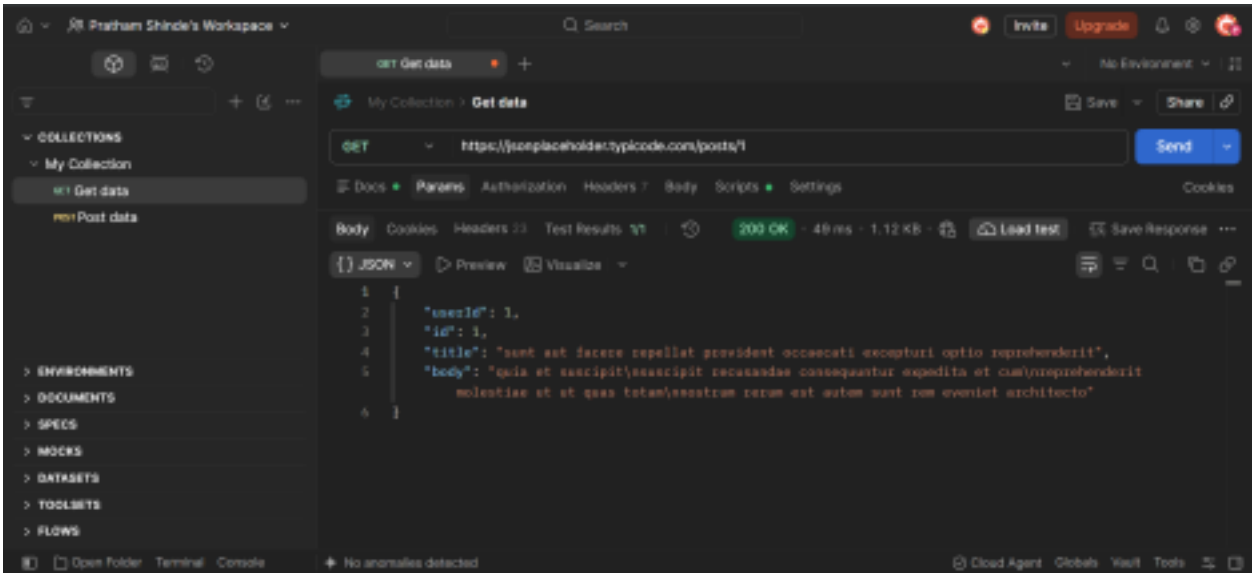
    message = wait.until(
        EC.visibility_of_element_located((By.ID, "message"))
    )

```

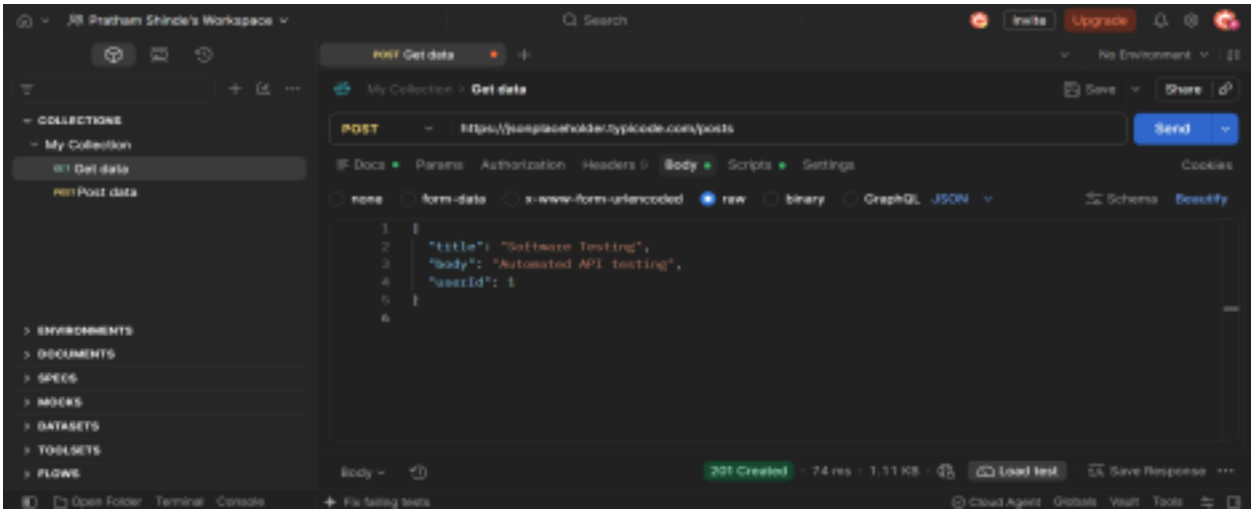
```
25
26      assert message.text == "Received!"
27
28      print("Test Passed")
29      print("Input entered successfully")
30      print("Form submitted successfully")
31
32  finally:
33      driver.quit()
34
```

B.2 Input and Output:

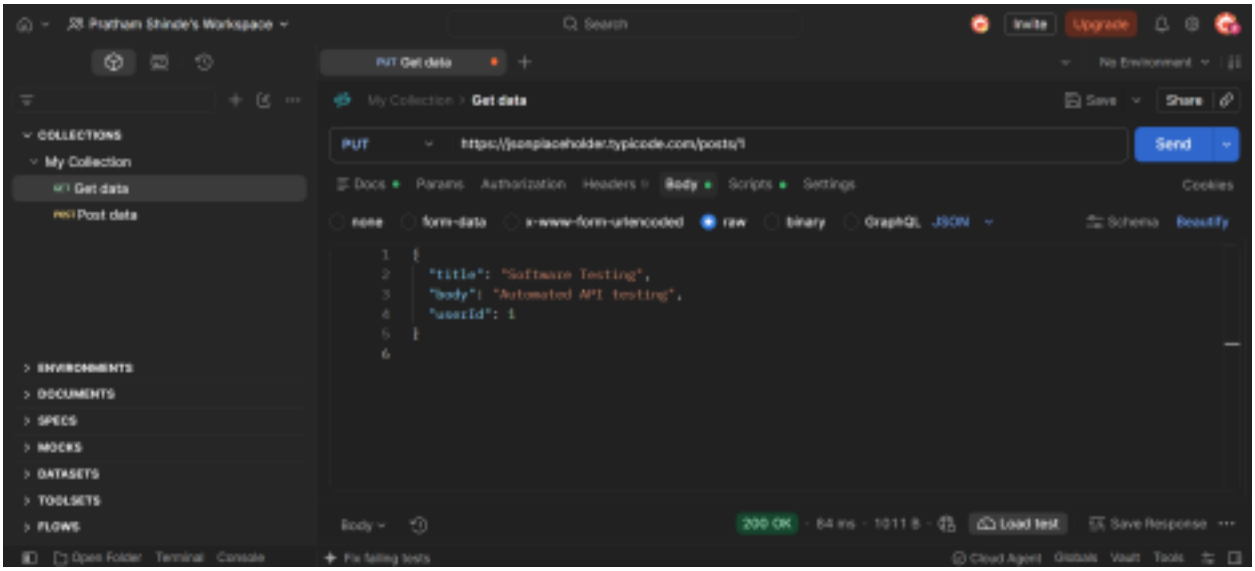
GET Output



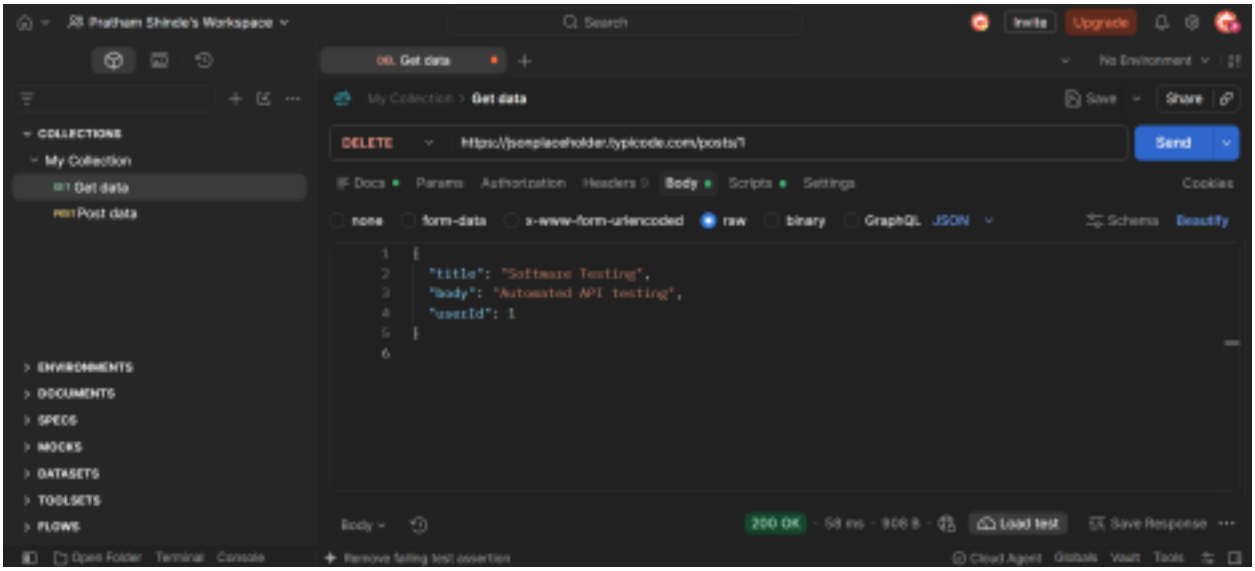
POST Output



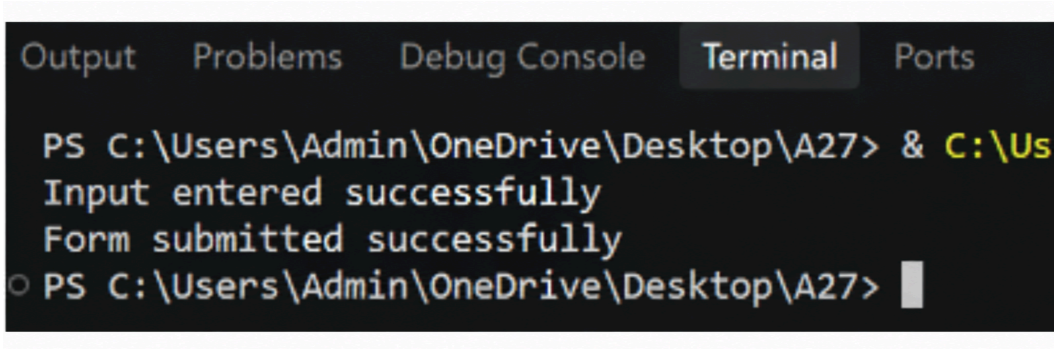
PUT Output



DELETE Output



Selenium Output



B.3 Conclusion:

The experiment was successfully performed to study automated software testing using Postman and Selenium WebDriver. API requests using GET, POST, PUT, and DELETE methods were executed in Postman, and the responses were examined using status codes and response data. Selenium WebDriver was used to automate web browser interactions and verify the results of a web form. This experiment demonstrated that automated testing reduces manual effort, improves accuracy and repeatability, saves execution time, and helps maintain software quality.

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